Grounded Chain-of-Evidence for Spatial Visual Question Answering

Wonmin Kim Hyun Woo Yu Anushka Purohit Tanishq Patil
University of California, San Diego
CSE 252D: Advanced Computer Vision (Spring 2026) – Project Proposal, Week 3
{wok016, hyy019, apurohit, t5patil}@ucsd.edu

1. Problem

Vision–Language Models (VLMs) such as LLaVA [6] frequently hallucinate on *relational* and *spatial* questions (e.g., "Is the red cup to the left of the blue plate?"), and agentic pipelines like ReAct [14] and ViperGPT [11] regenerate answers without verifying the visual evidence. Recent verification work such as ReCoVERR [10] recovers supporting evidence through follow-up VLM queries but does not explicitly verify geometric spatial relations—precisely where VLMs fail on spatial benchmarks [5].

Task. Given an RGB image and a question about a binary spatial predicate (`left_of`, `above`, `behind`, `on`, `contains`), output (i) a yes/no answer and (ii) a verified *Spatial Evidence Graph* recording which claims were checked and how. **Scope.** Single image, English questions, the predicates listed above. Counting, OCR, and free-form reasoning are out of scope. **Assumptions.** Off-the-shelf VLM, detector, and monocular depth model run locally; no fine-tuning.

2. Motivation

We propose a training-free agentic system that emits a **chain of grounded claims with explicit spatial–geometric verification**. Each claim is tied to bounding-box evidence and checked by a Critic using an open-vocabulary detector and rule-based geometric tests; on failure, the system crops the disputed region and re-examines it, or abstains. The *Spatial Evidence Graph* combines scene-graph structure with explicit geometric constraints—a domain-specific representation that reasons beyond raw pixels.

3. Key Background Literature

Verification. ReCoVERR [10] performs uncertainty-driven evidence recovery with selective abstention on existence/attribute claims; we generalize this line to geometric-relational claims. **Agentic VQA.** ViperGPT [11] and VisProg [3] synthesize programs without verification. **Active perception.** V* [12] and Visual CoT [9] crop and re-query without claim verification. **Spatial VLMs.** SpatialVLM [1] and SpatialRGPT [2] inject spatial reasoning via training; we are a complementary training-free wrapper. **Tools.** Grounding-DINO [7] and Depth Anything V2 [13].

4. Expected Outcomes

System. Three agents: *Planner* (LLM) parses the question into $\{\mathrm{obj}_1, \mathrm{obj}_2, \mathrm{rel}\}$; *Executor* (LLaVA-1.5, prompted to emit a JSON record with an answer and per-claim object names) produces hypotheses that Grounding-DINO localizes; *Critic* verifies each claim via a rule engine over bbox geometry and depth (x/y-centroid, IoU, median depth). The loop runs **error detection** (geometric mismatch), **correction** (crop and re-query), and **re-execution** (rebuild the graph), up to three rounds, with $k \in \{1, 2, 3\}$ swept as an ablation to characterize diminishing returns. Our contribution is (i) generalizing post-hoc verification from existence/attribute to geometric-relational claims and (ii) augmenting filter-or-keep verification with a crop-based active-perception fallback for failed claims.

Evaluation. *Datasets:* What's-Up [5], GQA [4] spatial subset, and 3DSRBench [8] for depth-involving relations. *Baselines:* raw LLaVA and LLaVA+ViperGPT (agentic-without- verification); SpatialRGPT [2] zero-shot as a trained-spatial-VLM reference; and an existence-grounding ablation of our own Critic (no geometric rules) as a verification-without- geometry reference. *Ablations:* no Critic, no crop fallback, 2D-only, and iteration sweep $k \in \{1, 2, 3\}$. *Metrics:* accuracy, selective accuracy (coverage–risk), claim-verification rate, average iterations; visualized evidence graphs and crop trajectories for successes and failures; a failure-mode taxonomy (detector miss, depth noise, VLM bias).

Deliverables & risks. A three-agent pipeline, 8-page CVPR-format report, and live demo. Principal risk: detector/depth noise causing false verification failures—mitigated by confidence thresholds, relative (not absolute) comparisons, and ablations. Secondary risk: insufficient improvement over existence-only verification—mitigated by a diagnostic set of relational hard-negatives where existence-only checks are provably insufficient.

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